

HDOC Day 38 - C Programming

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Batch: 15

Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

C Program

```
#include <stdio.h>

void countCharacters()
{
    char str[100];
    int i, spaces = 0, digits = 0, special = 0;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ' ')
            spaces++;
        else if (str[i] >= '0' && str[i] <= '9')
            digits++;
        else if (!((str[i] >= 'a' && str[i] <= 'z') ||
                    (str[i] >= 'A' && str[i] <= 'Z'))))
            special++;
    }

    printf("Number of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

void countFrequency()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    scanf("%[^\n]", str);
    printf("Enter the character to find: ");
    scanf("%c", &ch);

    for (i = 0; str[i] != '\0'; i++)
        if (str[i] == ch)
            count++;

    printf("Frequency of '%c' = %d\n", ch, count);
}

void toggleCase()
{
    char str[100];
    int i;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    for (i = 0; str[i] != '\0'; i++)
    {
```

```

        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
        else if (str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
    }

    printf("String after toggling case: %s\n", str);
}

int main()
{
    int choice;
    do
    {
        printf("\n----- STRING OPERATIONS MENU ----- \n");
        printf("1. Count spaces, digits and special characters\n");
        printf("2. Count frequency of a character\n");
        printf("3. Toggle case of each character\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: countCharacters(); break;
            case 2: countFrequency(); break;
            case 3: toggleCase(); break;
            case 4: printf("Exiting program...\n"); break;
            default: printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Execution Screenshot

```
Last login: Thu Oct 1 22:37:37 on ttys000
dhruvpandey@DHruVs-MacBook-Air ~ % nano code1.c
dhruvpandey@DHruVs-MacBook-Air ~ % clang code1.c -o code1
dhruvpandey@DHruVs-MacBook-Air ~ % ./code1
zsh: no such file or directory: ./code1
dhruvpandey@DHruVs-MacBook-Air ~ % ./code1.c
```

```
----- STRING OPERATIONS MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 1
Enter a string: Dhruv
Number of spaces = 0
Number of digits = 0
Number of special characters = 0
```

```
----- STRING OPERATIONS MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 2
Enter a string: Dhruv
Enter the character to find: D
Frequency of 'D' = 1
```

```
----- STRING OPERATIONS MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 3
Enter a string: Khyati
String after toggling case: kHYATI
```

```
----- STRING OPERATIONS MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 1
Enter a string: 
```